

**Dialogue through current turn
+ optional user profile**

Stage 1 · Candidate generation — full 47,071-track catalog

Lexical anchor

4 BM25 views + 4 exact match
RRF union (500 candidates)
anchor LightGBM · keep top 20

12 auxiliary sources

4 continuity · 3 query-neighbor
4 entity · 1 transition
up to 100 candidates each

Union · deduplicate · remove already-recommended
avg. 372 candidates per turn

Stage 2 · Learning-to-rank

73 rank-only features
anchor · source ranks
agreement · turn position

LightGBM LambdaRank
6 seeds · score averaging
trained on 8,000 dev turns

**Fixed top-20
ranking**

top-1 metadata

predicted_track_ids

Stage 3 · Grounded response generation

Hosted Claude Opus 4.8 · frozen
cannot change the ranking
inputs: dialogue + profile, top-1 track metadata
all 80 responses regenerated per submission

predicted_response